VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.

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Best of luck!

PAST PAPER

MTH601 Operations Research BC08040074

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

The cost matrix in assignment problem is always diagonal matrix

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Answer (Please select your correct option)

- square matrix



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- identity matrix
- zero matrix

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix

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Answer (Please select your correct option)

- Subtraction
- Division
- Multiplication
- Addition

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

During a replacement if the value of money decreases at the rate of **3**% then the present worth factor of unit amount to be spent after one year is given by

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Answer (Please select your correct option)

- 0.25
- 0.333
- 0.9708
- 4

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

In sequencing problems, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machines problems

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Answer (Please select your correct option)

- $3n$
- $3n!$
- $2 \times 3 = 6$
- 2

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

If the money carries a rate of interest of **12**% per year, the present worth factor of one rupee due in one year is

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Answer (Please select your correct option)

- 0.08333
- 0.89285 ⇦
- 0.0769
- 13

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Degeneracy in a 5×6 transportation problem occurs when the number of occupied cell is less than

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Answer (Please select your correct option)

- 20 but greater than 10
- 10
- Zero
- Infinity

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

We go in probabilistic replacement model when period between installation and failure is

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Answer (Please select your correct option)

- varying exponentially
- varying linearly
- Constant
- is not constant (varying arbitrarily)

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Any set of non negative allocations ($X_{ij} > 0$) which satisfies the row and column sum is called a _____ solution.

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Answer (Please select your correct option)

- feasible
- non basic feasible
- basic infeasible
- optimal

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

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Answer (Please select your correct option)

- Degenerate
- Multiple Solutions
- Non degenerate
- Alternative Optima

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following cost table:

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10

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Answer (Please select your correct option)

- 10
- 15
- 25
- 5

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Source	**S1**	5	1	2	**25**
	S2	7	2	4	**10**
	S3	3	3	5	**15**
	Demand	**10**	**20**	**20**	

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Answer (Please select your correct option)

- 10
- 15
- 25
- 5

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10

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Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only correct

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

	S3	3 10	3	5	15
	Demand	10	20	20	

Using Vogel's Approximation Method:
These below are the three cells which can be next cell:

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Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

I. S3 → D2
II. S2 → D2
III. S1 → D3

Which one is the correct option?

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Answer (Please select your correct option)

- I, II and III
- I only
- II only
- III only
- I and II only

NO IDEA

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

We go in probabilistic replacement model when period between installation and failure is

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Answer (Please select your correct option)

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- varying linearly
- Constant
- is not constant (varying arbitrarily)



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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If the mean arrival and mean service rates are **4** and 7 respectively in a queue then expected waiting time in the system is

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Answer (Please select your correct option)

- $\frac{1}{3}$
- 3
- 28
- $\frac{7}{4}$

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

In a bank, every 15 minutes one customer arrives for cashing the cheque. The staff in the payment counter takes only 10 minutes for serving a customer on an average, then the service rate " μ" = _____.

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Answer (Please select your correct option)

- 6 per hour
- 4 per hour
- 10 per hour
- 1/6 per hour

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

A duplicating machine maintained for office use is used and operated by people in the office who need to make copies. Since the work to be copied varies in length (number of pages of the original) and copies required, the service rate is randomly distributed, the arrival rate is 5 per hour and the service rate is 10 per hour then the equipment utilization " ρ " is equal to

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Answer (Please select your correct option)

- 0.50 ⇦ 0
- 0.20
- 5
- 2

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the average rate " λ " is

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Answer (Please select your correct option)

- 1/8 = 0.125
- 8
- 4
- ¼ = 0.25

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the mean service time " μ " is

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Answer (Please select your correct option)

- ½ = 0.5
- 4
- ¼ = 0.25
- 2

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

If "N_i" be the Number of replacement made at the end of the ith week and "P_j" be the probability of failure during the ith week, then N_1 = ----------.

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Answer (Please select your correct option)

- N_0P_1
- N_1P_1
- N_0P_0
- N_1P_2

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

The present worth of a rupee to be spent after a year is denoted by v and given by

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Answer (Please select your correct option)

- $v = 1/(1+r)$
- $v = (1+r)$
- $v = (1+r)/10$
- $v = (1/r)$

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Formula for a geometric series "$x + vx + v^2x + \ldots + v^{n-1}x$" is

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Answer (Please select your correct option)

- $P(n) = x\ \frac{1 - v^n}{1 - v}$
- $P(n) = \frac{1 - v^n}{1 - v}$
- $P(n) = x\frac{1 - v}{1 - v^n}$
- $P(n) = \frac{1 - v}{1 - v^n}$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Product of 'item cost' and 'ordered item' is

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Answer (Please select your correct option)

- Crash cost
- Cost period
- Set up cost
- Shortage cost

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Standard Deviation (S.D.) = _____ MAD

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Answer (Please select your correct option)

- $\sqrt{\frac{2\pi}{3}}$
- $\sqrt{\frac{2}{\pi}}$
- $\sqrt{\frac{\pi}{2}}$
- $\sqrt{\frac{3\pi}{2}}$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

MAD = --------- S.D.

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Answer (Please select your correct option)

- $\sqrt{\frac{2}{\pi}}$
- $\sqrt{\frac{\pi}{2}}$
- $\sqrt{\frac{2\pi}{3}}$
- $\sqrt{\frac{3\pi}{2}}$

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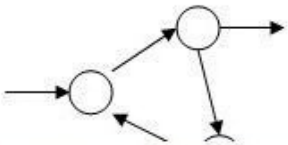
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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

The following network is an example of



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Answer (Please select your correct option)

- Redundancy
- Dangling
- Cycling
- Dummy

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

For any activity backward pass computations provide its

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Answer (Please select your correct option)

- Earliest start times
- Latest start times
- Moderate start times
- Completion time

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Best possible time estimate that a given activity would take under normal conditions which often exist, is called

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Answer (Please select your correct option)

- Most Likely time estimate
- Pessimistic time estimate
- Smallest time estimate
- Activity time estimate

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

In PERT, activity time estimates are distributed according to

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Answer (Please select your correct option)

- Beta Distribution
- Normal Distribution
- Poisson distribution
- Binomial Distribution

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Solution region for the constraint $y \leq 0$ is the

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Answer (Please select your correct option)

- Half plane below y-axis
- Half plane below the line: $y = 0$
- Set of all those points where ordinates are non-positive
- All are equivalent

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

In a linear Programming Problem (LPP), which of the following must be hold?

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Answer (Please select your correct option)

- Only objective function is linear
- Both objective function and constraints are linear
- Only constraints needs to be linear
- At least one of objective function or constraint should be linear

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If a company manufacture 'x' units of product 'A' and 'y' units of 'B' with associated profits of Rs.5 and Rs.3 then which of the following is the objective function to maximize is the profit?

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Answer (Please select your correct option)

- $z = 15xy$
- $z = 5x - 3y$
- $z = 3x - 5y$
- $z = 5x + 3y$

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The _____ variable is chosen by examining the cost coefficents in the objective function.

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Answer (Please select your correct option)

- entering
- leaving
- positive slack
- negative slack

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

While solving a linear programming problem by big M – Method, traditionally the _____ variables are chosen in the initial basic feasible solution.

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Answer (Please select your correct option)

- neagtive slack
- positive slack
- entering
- artificial

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

In the big-M method, if the introduced _____ variables do not leave the basis in the final iteration, then this indicates that the give linear programming problem cant be optimized.

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Answer (Please select your correct option)

- entering
- positive slack
- negative slack
- artificial

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Zero valued artificial variables may appear as _____ variables in the final solution, when one or more of the original constraints equations is redundant.

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Answer (Please select your correct option)

- non basic
- basic
- slacks
- surplus
- artificial

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

By using Two phase method to solve a linear programming problem, in phase I, a new objective function is formed by assigning on left hand side, zero to every original variable (including slack and surplus variables) and -------- to each of the artificial variables.

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Answer (Please select your correct option)

- M
- −M
- +1
- −1

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

For the linear programming problem;

$\text{Max}Z = 2x + 3y$

Subject to

$$\left.\begin{array}{l} x \geq 2 \\ y \leq 3 \\ x, y \geq 0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} x - s_1 + A = 2 \\ y + s_2 = 3 \\ x, y, s1, s2, A \geq 0 \end{array}\right\}$$

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Answer (Please select your correct option)

- $\text{Max}Z = 2x + 3y + A$
- $\text{Min}Z = 2x + 3y + A$
- $\text{Max}Z = A$
- $\text{Min}Z = A$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

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Answer (Please select your correct option)

- $\text{Max}Z = 2x + 3y + A$
- $\text{Min}Z = 2x + 3y + A$
- $\text{Max}Z = A$
- $\text{Min}Z = A$ ⇦

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

$y \leq 3$, $x, y \geq 0$ $\Rightarrow y + s_2 = 3$, $x, y, s1, s2, A \geq 0$

Which of the following is associated objective function of the 1st phase?

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Answer (Please select your correct option)

- $MaxZ = 2x + 3y + A$
- $MinZ = 2x + 3y + A$
- $MaxZ = A$
- $MinZ = A$

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

The inequality $2x + 3y \geq 18$ is equivalent to

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Answer (Please select your correct option)

- $2x + 3y \leq -18$
- $-2x - 3y \geq 18$
- $-2x - 3y \geq -18$
- $-2x - 3y \leq -18$ ⇦

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MTH601 Operations Research BC08040074

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

In which of the following models, Simplex algorithm is not preferred to use due to laborious computations?

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Answer (Please select your correct option)

- Transportations models
- Degenerate Linear models
- Non-degenerate Linear models
- Dual or unbounded linear models

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Transportations models consist of ------- like the production centers and --------- which may be the sales centers.

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Answer (Please select your correct option)

- (sinks, sources)
- (sources, sinks)
- (origins, sources)
- (sinks, destinations)

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

To convert the transportation problem into a maximization model we have to----------.

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Answer (Please select your correct option)

- write the inverse of the coefficient matrix
- multiply the feasibility condition by −1
- multiply the coefficient matrix by −1
- We can't convert the transportation problem into a maximization problem, as it is basically a minimization prob

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